Pricing Toolbox: A Guided Refresher

Lecture 14 Notes — Itô, Girsanov, Path Dependence, and Early Exercise

The four questions

Question	Tool
How do we simplify a nonlinear Brownian quantity?	Itô formula
How do we remove a drift?	Girsanov
What information must a payoff remember?	state augmentation
When should the holder stop?	dynamic programming

The methods change a representation while preserving the pricing problem.

1 Itô calculus turns integrals into identities

For $f(t, B_t)$,

$$\mathrm{d}f(t, B_t) = f_t\,\mathrm{d}t + f_x\,\mathrm{d}B_t + \frac{1}{2}f_{xx}\,\mathrm{d}t.$$

Taking $f(x) = e^x$ gives

$$\mathrm{d}e^{B_t} = e^{B_t}\,\mathrm{d}B_t + \frac{1}{2}e^{B_t}\,\mathrm{d}t,$$

hence

$$\boxed{\int_0^T e^{B_s}\,\mathrm{d}B_s = e^{B_T} - 1 - \frac{1}{2}\int_0^T e^{B_s}\,\mathrm{d}s.}$$

Example — the stochastic exponential

$$\mathcal{E}_t(\theta) = \exp\left(\theta B_t - \frac{1}{2}\theta^2 t\right)$$

satisfies $\mathrm{d}\mathcal{E}_t = \theta\mathcal{E}_t\,\mathrm{d}B_t$. The compensating quadratic-variation term is why it is a martingale rather than merely an exponential function of Brownian motion.

Exercise — Itô correction

Why is $\exp(\theta B_t)$ not a martingale for $\theta \neq 0$?

Short solution

Itô gives a positive drift $\frac{1}{2}\theta^2\exp(\theta B_t)\,\mathrm{d}t$. Subtracting $\frac{1}{2}\theta^2 t$ in the exponent cancels that drift.

2 Girsanov removes drift by changing measure

If $W_t^{\mathbb{Q}} = W_t + \int_0^t \lambda_s\,\mathrm{d}s$ is Brownian motion under a new measure $\mathbb{Q}$, then a process with drift under the old measure can become a driftless Brownian motion under $\mathbb{Q}$. The density is a stochastic exponential, subject to an integrability condition.

The purpose of the change

Changing measure reweights paths; it does not change observed path values. For pricing, choose the measure under which discounted traded assets are martingales. For hitting problems, choose the measure under which the underlying Brownian calculation is simple.

Example — first hitting time

For $X_t = \mu t + B_t$, the event of hitting level $a > 0$ can be related by Girsanov to a driftless Brownian hitting event. Reflection then yields the inverse-Gaussian density

$$f_{\tau_a}(t) = \frac{a}{\sqrt{2\pi t^3}} \exp\left(-\frac{(a-\mu t)^2}{2t}\right), \qquad t > 0.$$

3 Path-dependent claims need an enlarged state

For an arithmetic-average Asian call,

$$\left(\frac{1}{T}\int_0^T S_u \, \mathrm{d}u - K\right)^+,$$

define

$$I_t = \int_0^t S_u \, \mathrm{d}u, \qquad \mathrm{d}I_t = S_t \, \mathrm{d}t.$$

The stock alone is not Markov for this payoff; the state is (t, S_t, I_t). Under Black–Scholes, the value solves

$$v_t + rsv_s + sv_i + \frac{1}{2}\sigma^2 s^2 v_{ss} - rv = 0,$$

with terminal payoff $(I/T - K)^+$.

Simulation has two errors here

A Monte Carlo Asian estimator has sampling error and numerical integration error for the time integral. A geometric Asian option is often a useful control variate because its price is tractable while it is strongly correlated with the arithmetic payoff.

Exercise — a barrier-monitoring bias

Why can a discretely monitored up-and-in option be underpriced relative to continuous monitoring?

Short solution

A path may cross the barrier between observation dates and return below it before the next observation. The discrete estimator misses that activation; refining the grid reduces the bias.

4 Barrier events combine maxima and measure changes

For an up-and-in one-touch with barrier $H > S_0$, the payoff is

$$\mathbf{1}_{\{\sup_{0\le u\le T} S_u \ge H\}}.$$

With $X_t = \log(S_t/S_0) = \mu t + \sigma B_t$, the barrier becomes $a = \log(H/S_0)$. Girsanov handles drift and reflection handles the maximum. The price is the discounted hitting probability.

5 American options are stopping decisions

In a binomial model with gross return R and risk-neutral probability p,

$$\boxed{V_t = \max\left\{g(S_t), \frac{pV_{t+1}^u + (1-p)V_{t+1}^d}{R}\right\}.}$$

The first term is exercise value and the second is continuation value.

Example — a three-step American put

Let $S_0 = 100$, $u = 1.1$, $d = .9$, $R = 1.02$, $K = 100$. Then $p = .6$. At node $S_2 = 81$,

$$\text{exercise} = 19, \qquad \text{continuation} = \frac{.6(10.9) + .4(27.1)}{1.02} = 17.04.$$

Exercise is optimal. Working backward gives approximately

$$V_0^{\text{Eur}} = 4.59, \qquad V_0^{\text{Am}} = 4.91.$$

The difference is the early-exercise premium.

Exercise — the correct operation

At a node, a put has exercise value 8 and continuation value 9.2. What is its American value and action?

Short solution

The value is $\max\{8, 9.2\} = 9.2$. Continue; exercise and continuation are alternative policies, not cash flows to add.

6 Numerical habits complete the toolbox

First diagnostic by contract feature

Problem	First check
Asian Monte Carlo	integration grid and sampling error
Barrier simulation	monitoring bias and bridge correction
American tree	time-step refinement and exercise region
PDE or local-volatility grid	boundary placement and mesh convergence

A number is not yet a result

Before trusting a numerical price, apply a relevant refinement or no-arbitrage check. A formula can be correct while its discretization, simulation, or implementation is not.

Final synthesis

The payoff tells us what information must be carried forward: quadratic variation, a probability density, a state variable, a path maximum, or a stopping rule.